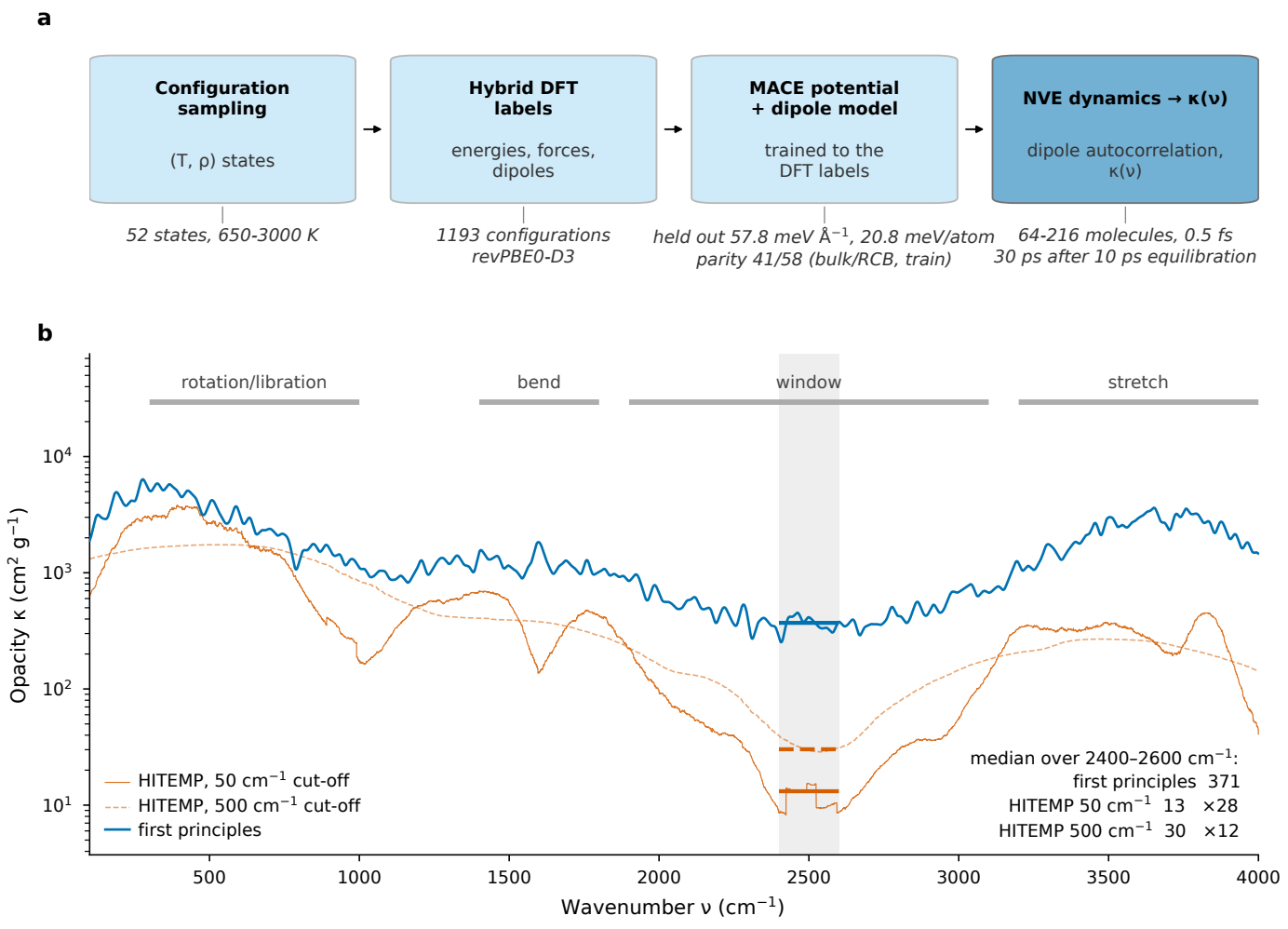

a
Configuration sampling
(T, ρ) states
Hybrid DFT labels
energies, forces, dipoles
MACE potential + dipole model
trained to the DFT labels
NVE dynamics → κ(ν)
dipole autocorrelation, κ(ν)
52 states, 650-3000 K
1193 configurations
revPBE0-D3
held out 57.8 meV Å^{-1}, 20.8 meV/atom
parity 41/58 (bulk/RCB, train)
64-216 molecules, 0.5 fs
30 ps after 10 ps equilibration
b
rotation/libration
bend
window
stretch
Opacity κ (cm^2 g^{-1})
Wavenumber ν (cm^{-1})
HITEMP, 50 cm^{-1} cut-off
HITEMP, 500 cm^{-1} cut-off
first principles
median over 2400–2600 cm^{-1}:
first principles 371
HITEMP 50 cm^{-1} 13 ×28
HITEMP 500 cm^{-1} 30 ×12